

THE REMAINING RESIDUE CLASS OF THE CUBIC BORWEIN CONJECTURE: A COMPUTER-ASSISTED PROOF

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ABSTRACT. Let $P_n(q) = \prod_{k \leq 3n, 3 \nmid k} (1 - q^k)$. The cubic Borwein conjecture of Krattenthaler predicts that the coefficients of $P_n(q)^3$ have the sign pattern $+-$ according to the exponent modulo 3. Krattenthaler and Wang proved two thirds of it, leaving open the non-positivity of the coefficients of q^{3M+2} in the lower half of the range (equivalently, of q^{3M+1} in the upper half). We prove this remaining case, in the strong form $[q^m]P_n(q)^3 < 0$ for $3n \leq m \leq \frac{9}{2}n^2$, $m \equiv 2 \pmod{3}$.

The proof is computer-assisted. For $n \leq 243$ the claim is verified by exact integer arithmetic. For $n \geq 244$ we use a reduction in which the class-2 coefficient becomes a Laplace integral with positive main term, a band lemma just above the cutoff, and a complex saddle-point analysis of P_n^3 near the centre of the range. Every quantitative inequality needed for $n \geq 244$ is established *uniformly in n* by explicit ball-arithmetic certificates, whose parameter domains are shown to cover all admissible (n, m) . All code and logs are public.

1. INTRODUCTION

1.1. The conjectures. Write $P_n(q) = \prod_{k \leq 3n, 3 \nmid k} (1 - q^k) = (q; q)_{3n} / (q^3; q^3)_n$ and $c_n(m) = [q^m]P_n(q)^3$. The polynomial P_n^3 has degree $9n^2$ and is palindromic, so it suffices to consider $m \leq \frac{9}{2}n^2$. Peter Borwein conjectured that P_n has the sign pattern $+-$ (the First Borwein Conjecture), and that the same holds for P_n^2 (the Second). Andrews [1] obtained partial results. The First Borwein Conjecture was proved by Wang [5]. Krattenthaler and Wang [4] developed an asymptotic framework which gave a new proof of the first conjecture, a proof of the second, and a proof of “two thirds” of the following cubic analogue, due to Krattenthaler.

Conjecture 1.1 (cubic Borwein conjecture). For $n \geq 1$ and $0 \leq m \leq \frac{9}{2}n^2$: $c_n(m) \geq 0$ if $m \equiv 0$, and $c_n(m) \leq 0$ if $m \equiv 1, 2 \pmod{3}$.

The case left open in [4, §11] is non-positivity of $c_n(3M+2)$ for $3M+2 \leq \frac{9}{2}n^2$ (and, by palindromy, of the coefficients of q^{3M+1} in the upper half).

Theorem 1.2. For all $n \geq 1$ and all $m \equiv 2 \pmod{3}$ with $m \leq \frac{9}{2}n^2$: $c_n(m) = 0$ if $m < 3n$, and $c_n(m) < 0$ if $3n \leq m \leq \frac{9}{2}n^2$.

Together with [4] this proves Conjecture 1.1.

1.2. Why the class $m \equiv 2$ is different. Put $P_\infty(q) = \prod_{3 \nmid k} (1 - q^k)$ and $E_n(q) = P_\infty^3 / P_n^3 = \prod_{k > 3n, 3 \nmid k} (1 - q^k)^{-3}$. Since $P_\infty^3 = (q; q)_\infty^3 / (q^3; q^3)_\infty^3$ and Jacobi’s identity $(q; q)_\infty^3 = \sum_{j \geq 0} (-1)^j (2j+1) q^{j(j+1)/2}$ involves only exponents $\equiv 0, 1 \pmod{3}$, the class-2 coefficients of P_∞^3 vanish identically. Hence $c_n(3M+2)$ is produced entirely by the tail E_n , and at the dominant cusp the

Date: September 17, 2026.

2020 Mathematics Subject Classification. Primary 11P82; Secondary 05A17, 41A60, 65G20.

Key words and phrases. Borwein conjecture, sign patterns, circle method, saddle point method, interval arithmetic, computer-assisted proof.

leading saddle-point contribution cancels. The class-2 coefficient is exponentially smaller than the terms it is built from, and the first-order analysis of [4] does not determine its sign.

1.3. Strategy. Let $\omega = e^{2\pi i/3}$, $a = \pi^2/9$, $N = 3n + 1$. The proof has four parts.

- (1) *Exact computation* for $n \leq 243$ (§3).
- (2) *Band lemma* for $3n + 2 \leq m < 6n + 2$ (§4): here $c_n(m) = 3 \sum_{i \leq M-n} (p_3(3i) + p_3(3i+1))$ and each summand is negative.
- (3) *Laplace regime* (§5). Writing $\log E_n(\omega q) = S + i\Delta$, one has $0 < \Delta < \pi/2$ on $(0, 1)$ (Lemma 2.1), so the phase $\hat{R}(w) = a/w + (m - \frac{1}{4})w + S + \log \sin \Delta$ is real on the positive axis and the class-2 coefficient becomes a real Laplace integral with positive main term (Proposition 2.5). We bound it uniformly in n for $\tilde{\mu} := (m - N - \frac{1}{4})/N^2 < 0.0985$.
- (4) *Centre regime* (§6): for $\mu := m/N^2 \geq 0.098$ we work with $P_n(q)^3$ directly at complex saddles near $q = r\omega^{\pm 1}$, again uniformly in n .

1.4. The role of the computer. The computer is used in two ways: an exact integer computation (§3), and the rigorous enclosure, in midpoint-radius (ball) arithmetic [3], of explicitly stated real inequalities on explicit coverings of compact parameter domains (§§4–6). No floating-point value enters a proof step. Uniformity in n is obtained by treating $\epsilon = 1/N$ as an interval variable containing 0, together with explicit monotonicity statements. Each certificate is a named script; §8 lists them with what they certify, and §9 lists independent consistency checks that are *not* part of the proof.

2. STRUCTURAL LEMMAS AND THE REDUCTION

Throughout $h(y) = \log(1 + e^{-y} + e^{-2y})$, $\Phi(v) = \int_v^\infty h = \text{Li}_2(e^{-v}) - \frac{1}{3} \text{Li}_2(e^{-3v})$, $\Psi(v) = 2 \text{Li}_2(e^{-v})$, and $\Phi(0) = a$.

Lemma 2.1 (tail phase). *For $0 < q < 1$: $\text{Im} \log E_n(\omega q) = 3 \sum_{A \geq n} [f(q^{3A+1}) - f(q^{3A+2})] \in (0, \pi/2)$, where $f(x) = \arctan(\sqrt{3}x/(2+x))$.*

Proof. $\log E_n(\omega q) = -3 \sum_{k > 3n, 3 \nmid k} \log(1 - \omega^k q^k)$ and $\text{Im} \log(1 - \omega x) = -f(x)$ for $x \in (0, 1)$, with the sign reversed for $\bar{\omega}$. Each bracket is positive since f is increasing. Also $f(q^{3A+1}) - f(q^{3A+2}) \leq f(q^{3A+1}) - f(q^{3A+4})$, which telescopes to at most $f(q^{3n+1}) < \pi/6$. $\square$

Lemma 2.2 (closed forms). *For $|q| < 1$, with $\chi'(r) = \pm 1$ for $r \equiv \pm 1 \pmod{3}$,*

$$\log E_n(\omega q) = S(q) + i\Delta(q), \quad S = \sum_{3 \nmid r} \frac{-3}{2r} \frac{q^{Nr} + q^{(N+1)r}}{1 - q^{3r}} + \sum_{3 \mid r} \frac{3}{r} \frac{q^{Nr} + q^{(N+1)r}}{1 - q^{3r}}, \quad \Delta = \sum_{3 \nmid r} \frac{3\sqrt{3}}{2r} \chi'(r) \frac{q^{Nr}(1 - q^r)}{1 - q^{3r}},$$

and $\text{Im} E_n(\omega q) = e^S \sin \Delta$ as holomorphic functions. For real q , S, Δ are real.

Proof. Expand $-3 \log(1 - \omega^k q^k) = 3 \sum_{r \geq 1} \omega^{kr} q^{kr} / r$ and sum over $k = 3A + 1, 3A + 2$ with $A \geq n$; then $\omega^r, \omega^{2r} \in \{1, \omega, \bar{\omega}\}$ gives the real and imaginary parts termwise. Both sides are holomorphic in $|q| < 1$. $\square$

2.1. The expansion of p_3 . Let $p_3(i) = [q^i] P_\infty(q)^3$, $B = \frac{1}{2}$, and for $k \geq 1$

$$\mathcal{P}_k(i) = \frac{2\pi}{k} \sqrt{\frac{B}{2i - B}} I_1\left(\frac{2\pi}{k} \sqrt{B(2i - B)}\right), \quad \mathcal{P}_3(i) = \sqrt{\frac{a}{i - \frac{1}{4}}} I_1\left(2\sqrt{a(i - \frac{1}{4})}\right).$$

Let $\chi(i) = 1, -1, 0$ for $i \equiv 0, 1, 2 \pmod{3}$.

Proposition 2.3 (explicit expansion; Appendix A). *For every $i \geq 1$,*

$$p_3(i) = \sqrt{3}\chi(i)\mathcal{P}_3(i) + \rho(i), \quad |\rho(i)| \leq \sum_{\substack{3|k \leq K(i) \\ k \geq 6}} \varphi(k)\mathcal{P}_k(i) + E_{\text{con}}, \quad K(i) = \lceil \sqrt{4\pi i} \rceil + 1,$$

with $E_{\text{con}} = 50.1501$.

Lemma 2.4 (Poisson form). $\mathcal{P}(q) := \sum_{i \geq 1} \mathcal{P}_3(i)q^i$ satisfies, for $q = e^{-w}$, $\text{Re } w > 0$, $|\text{Im } w| \leq \pi$,

$$\mathcal{P}(e^{-w}) = e^{-w/4}(e^{a/w} + \eta(w)), \quad \eta(w) = -1 + \sum_{k \neq 0} (-i)^k (e^{a/(w+2\pi ik)} - 1), \quad |\eta(w)| \leq 2.$$

Proof. With $F(x) = \sqrt{a/x} I_1(2\sqrt{ax})$ for $x > 0$ and $F = 0$ for $x \leq 0$ we have $\int_0^\infty F(x)e^{-wx}dx = e^{a/w} - 1$. Poisson summation over the shifted lattice $x = i - \frac{1}{4}$ gives the formula. For the bound write $e^u - 1 = u + r(u)$; $\sum_{k \neq 0} |r| \leq \frac{a^2}{6} e^{a/\pi} < 0.285$, and the linear part, grouped in blocks of four consecutive k , is at most 0.714. A ball-arithmetic enclosure over $\text{Re } w \in (0, 10]$, $|\text{Im } w| \leq \pi$ gives $|\eta| \leq 1.42$ (`poisson_check.py`). $\square$

Proposition 2.5 (reduction). *Let $m \equiv 2 \pmod{3}$ and let $e(j) \geq 0$ be the coefficients of E_n . Then*

$$c_n(m) = L(m) + R(m), \quad L(m) = -2 \text{Im} [q^m] \mathcal{P}(q) E_n(\omega q), \quad R(m) = \sum_j e(j) \rho(m-j),$$

and for every $W > 0$,

$$|R(m)| \leq \sum_{\substack{3|k \leq K(m) \\ k \geq 6}} 3\varphi(k) e^{mW+a_k/W} E_n(e^{-W}) + E_{\text{con}} e^{mW} E_n(e^{-W}), \quad a_k = a(3/k)^2.$$

Moreover $[q^m] \mathcal{P}(q) E_n(\omega q) = J_0 + J_\eta$ with, on the line $w = W + iy$,

$$J_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{a/w+(m-\frac{1}{4})w} E_n(\omega e^{-w}) dy, \quad J_\eta = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{(m-\frac{1}{4})w} \eta(w) E_n(\omega e^{-w}) dy,$$

and $\text{Im } J_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \text{Re } e^{\hat{R}(W+iy)} dy$ with $\hat{R}(w) = a/w + (m - \frac{1}{4})w + S(e^{-w}) + \log \sin \Delta(e^{-w})$. In particular $c_n(m) < 0$ whenever

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \text{Re } e^{\hat{R}(W+iy)} dy > \frac{1}{2\pi} \int_{-\pi}^{\pi} 3 e^{(m-\frac{1}{4})W} |\text{Im } E_n(\omega e^{-W-iy})| dy + \frac{|R(m)|}{2}. \quad (1)$$

Proof. $c_n(m) = \sum_j e(j)p_3(m-j)$. Insert Proposition 2.3. Since $\sqrt{3}\chi(i) = -2 \text{Im } \omega^{-i} \cdot (-1)^i$ is encoded by the identity $\sum_j e(j)\sqrt{3}\chi(m-j)\mathcal{P}_3(m-j) = -2 \text{Im} \sum_j e(j)\omega^j \mathcal{P}_3(m-j)$ for $m \equiv 2 \pmod{3}$ (check the three residues of j), the main part is $L(m)$. For $R(m)$ use $e(j) \geq 0$, $\mathcal{P}_k \geq 0$, $|\text{amplitude}| \leq \varphi(k)$, Cauchy's estimate on $|q| = e^{-W}$, and $\sum_i \mathcal{P}_k(i)e^{-Wi} \leq (1 + 2a_k)e^{a_k/W} \leq 3e^{a_k/W}$ (from $I_1(y) \leq \frac{y}{2}e^y$ and the Laplace transform). The integral formula is Cauchy's formula with Lemma 2.4; the coefficients of $\mathcal{P} \cdot \text{Im } E_n(\omega \cdot)$ are real, so $\text{Im } J$ equals the integral of the real part of $e^{a/w+(m-\frac{1}{4})w} \text{Im } E_n(\omega e^{-w}) = e^{\hat{R}}$ (Lemma 2.2), plus the η -part bounded by $|\eta| \leq 3$ in modulus. Finally $\text{Im } J > |R|/2$ gives $c_n(m) = -2 \text{Im } J + R < 0$. $\square$

Remark 2.6. On the real axis $\sin \Delta > 0$ by Lemma 2.1, so $\hat{R}$ is real there. We always take W to be the real saddle of $\hat{R}$. Writing $\log \sin \Delta = \log Q - Nw + \log(\sin \Delta / \Delta)$ with $Q = \Delta e^{Nw}$, the saddle equation reads $-a/W^2 + (m - N - \frac{1}{4}) + \partial_w(S + \log Q + \log \frac{\sin \Delta}{\Delta}) = 0$: the effective parameter is $\tilde{\mu} = (m - N - \frac{1}{4})/N^2$, not m/N^2 .

3. THE FINITE RANGE $n \leq 243$

Certificate 3.1 (`exact/cubic_check.c`). For $1 \leq n \leq 243$ and every $m \equiv 2 \pmod{3}$, $m \leq \frac{9}{2}n^2$: $c_n(m) = 0$ for $m < 3n$ and $c_n(m) < 0$ for $3n \leq m$.

The program multiplies out P_n^3 in exact GMP integers, applying $(1 - q^k)^3 = 1 - 3q^k + 3q^{2k} - q^{3k}$ in one descending pass per k , and checks the sign of every class-2 coefficient.

4. THE BAND $3n + 2 \leq m < 6n + 2$

Proposition 4.1. Let $n \geq 1$, $m = 3M + 2$ with $3n + 2 \leq m < 6n + 2$. Then $c_n(m) = 3 \sum_{i=0}^{M-n} g(i)$ with $g(i) = p_3(3i) + p_3(3i + 1)$, and $g(i) < 0$ for all $i \geq 0$. Hence $c_n(m) < 0$.

Proof. For $j < 2N$ the coefficient $e(j)$ equals 1 at $j = 0$, 3 for $N \leq j < 2N$ with $3 \nmid j$, and 0 otherwise. Since $p_3(m) = 0$, $c_n(m) = 3 \sum_{N \leq K \leq m, 3 \nmid K} p_3(m - K)$; for $K = 3A + 1$ and $K = 3A + 2$ ($n \leq A \leq M$) the arguments are $3(M - A) + 1$ and $3(M - A)$. The negativity of g is Certificate 4.2. $\square$

Certificate 4.2 (`band_certificate.py`). (a) $g(i) < 0$ for $0 \leq i \leq 2000$, by exact integers (p_3 from Jacobi's identity and the recurrence for $1/(q; q)_\infty^3$). (b) For $i > 2000$, by Proposition 2.3, $-g(i) \geq \sqrt{3}(\mathcal{P}_3(3i + 1) - \mathcal{P}_3(3i)) - |\rho(3i)| - |\rho(3i + 1)|$, with $\mathcal{P}_3(j + 1) - \mathcal{P}_3(j) \geq aI_2(2\sqrt{a(j - \frac{1}{4})})/(j + \frac{3}{4})$ and the envelopes $\kappa(x_0)e^x/\sqrt{2\pi x} \leq I_\nu(x) \leq e^x/\sqrt{2\pi x}$ ($x \geq x_0$). The resulting ratio is a finite sum of terms $Cj^pe^{-g\sqrt{j}}$ with $p \leq 3$, $g \geq 2\pi/9$, hence decreasing for $j \geq 74$, and at $j = 6000$ it is below 10^{-33} .

5. THE LAPLACE REGIME $\tilde{\mu} < 0.0985$

Here $n \geq 244$ and $m \geq 2N$. We verify (1) uniformly in $N \geq 733$.

5.1. Scaled variables. Put $\epsilon = 1/N$ and $\zeta = Nw$. With $g(x) = x/(1 - e^{-x})$ and $h(x) = (1 - e^{-x})/(1 - e^{-3x}) = g(3x)/(3g(x))$, Lemma 2.2 gives $S = N\tilde{S}(\zeta, \epsilon)$ and $Q = \tilde{Q}(\zeta, \epsilon)$ where

$$\tilde{S} = \sum_r s_r e^{-r\zeta} (1 + e^{-r\zeta\epsilon}) \frac{g(3r\zeta\epsilon)}{3r\zeta}, \quad \tilde{Q} = \sum_{3 \nmid r} \frac{3\sqrt{3}\chi'(r)}{2r} e^{-(r-1)\zeta} h(r\zeta\epsilon),$$

$s_r = -\frac{3}{2r}$ ($3 \nmid r$), $\frac{3}{r}$ ($3 \mid r$). Both are analytic in ϵ near $\epsilon = 0$ because g is (its Taylor coefficients are the Bernoulli numbers; we use the truncated series with the remainder bound $|b_k| \leq 4(2\pi)^{-k}$). Then

$$\hat{R}(w) = N G(\zeta) + \frac{(m - N - \frac{1}{4})\zeta}{N}, \quad G(\zeta) = \frac{a}{\zeta} + \tilde{S} + \epsilon(\log \tilde{Q} + \log \frac{\sin \Delta}{\Delta}),$$

and the real saddle $v = NW$ satisfies $-G'(v) = \tilde{\mu}$. We parametrise by the pair (v, ϵ) ; a ball of pairs covers all (N, m) whose saddle and $1/N$ lie in it.

5.2. The certificate for one ball. Fix a ball $V \ni v$, $E \ni \epsilon$, $E \subset [0, 1/733]$, and let $N_b = \max(733, 1/\sup E)$ be the smallest N it contains. Write $\sigma_y = N^{-3/2}G''(v)^{-1/2}$ and $\tau = y/\sigma_y$. The certificate `B_uniform.py` computes, in ball arithmetic:

- (1) *Window* $|\tau| \leq c$ ($c \in \{2, 2.5, 3\}$). Taylor expansion of G at v to order four with coefficients $G_3 = \sqrt{\epsilon}|G'''|/G''^{3/2}$, $\phi_4 = \epsilon G''''/G''^2$, and a fifth-order remainder bounded by Cauchy's estimate on the circle $|\zeta - v| = 0.6v$ (the a/ζ part exactly). This gives a pointwise lower bound $L(\tau)$ for $\operatorname{Re} e^{\hat{R} - \hat{R}(W)}$ and the lower bound $\sigma_y \int_{-c}^c L$.

- (2) *Zone 1*, $\tau \in [c, \tau_s]$ with $t_s = N_b \sigma_y \tau_s$ fixed: $\widehat{R}(W + iy) - \widehat{R}(W) \leq -\kappa \tau^2/2$, where κ comes from even-order Taylor terms and Cauchy bounds for the sixth derivative of $a/\zeta + \widetilde{S}$ and of $\log \widetilde{Q} + \log \frac{\sin \Delta}{\Delta}$; κ is non-decreasing in N .
- (3) *Zone 2*, $t = Ny \in [t_s, t_C]$: exact enclosures of $\operatorname{Re}[\widehat{R}(W + iy) - \widehat{R}(W)] = N A(t) + B(t)$ on cells, with the assertion $A < -1/(2N_b)$ on every cell. Then $\sqrt{N} e^{NA+B}$ is non-increasing in $N \geq N_b$.
- (4) *Zone 3*, $t \geq t_C$: either $|\operatorname{Im} E_n(\omega q)| \leq E_n(|q|)$ and $|e^{a/w}| \leq e^{Nav/(v^2+t_C^2)}$, with a negative exponent slope below $-3/(2N_b)$; or, if $\inf E > 0$, $|\widetilde{Q}| \leq \sum c_r e^{-(r-1)v} \cdot 2(1/(3rv \inf E) + 1)$ and $|\Delta| \leq |\widetilde{Q}| e^{-v}$.
- (5) *η -part*: the integrand of (1) is $3e^{\operatorname{Re}(\widehat{R}-\widehat{R}(W))-\operatorname{Re}(a/w)}$ relative to $e^{\widehat{R}(W)}$; bounded on the same cells.
- (6) *$R(m)$* : Proposition 2.5 with $W' = tW$ for t in a finite set, using $\log E_n(e^{-W'}) \leq N\Psi(v')/v' + 6|\log(1 - e^{-v'})|$ and $\tilde{\mu} = -G'(v)$; the exponent slope in N is asserted below $-7/(2N_b)$.

The ball passes if (zone 1 + zone 2 + zone 3 + η -part + R -part)/(window lower bound) < 1 , which is (1) for every (N, m) in the ball.

Pairs with $\epsilon > \tilde{\mu}/0.999$ do not occur (since $m \geq 2N$ gives $\tilde{\mu} \geq \epsilon(1 - \frac{1}{4N})$); such balls are discarded only after an enclosure $\tilde{\mu} \leq a/v^2 + \max |h'|$ (Cauchy on $|\zeta - v| = v/2$) proves it. Balls with $\tilde{\mu} > 0.0985$ belong to §6.

Certificate 5.1 (`run_uniform.py`, 36 chunks). Every ball of an adaptive covering of $v \in [2.95, 3000]$, $\epsilon \in [0, 1/733]$ passes; the worst ratio is 0.117. Moreover $-G'(2.95) \geq 0.1015$ for all $\epsilon \in [0, 1/733]$ and $G'' > 0$ on every ball, so every saddle with $\tilde{\mu} < 0.0985$ has $v > 2.95$.

Proposition 5.2 (closing lemma; `ray_asymptotic2.py`). For $v \geq 3000$ and all $N \geq 733$, (1) holds with ratio at most 0.0082.

Sketch. Put $s = ev^2/a \in (0, 1.0011]$. On $|\zeta - v| \leq v/2$ one has $|\widetilde{S}| \leq 5 \cdot 10^{-655}$ and $|h(x)| \leq 4$ for all ϵ (for $|x| \geq 0.9$ because $|x|/\operatorname{Re} x = |\zeta|/\operatorname{Re} \zeta \leq 3$), so $|\Delta| \leq 4 \cdot 10^{-651}$ and the relative perturbations of the derivatives of G from a/ζ are at most $4.7 \cdot 10^{-5}$. The window then has $G_3 \leq 0.0388$, $\phi_4 \leq 2 \cdot 10^{-3}$, $G_5 \leq 10^{-3}$ and lower bound 2.494; the Gaussian tail is at most 0.0205; for $|t| \geq v/2$ the exponent is at most $-\frac{v}{5s} + \log(\frac{4v}{3as} + 6) + \frac{20ve^{-v}}{as}$, whose derivative in s is positive on $(0, 1.0011]$ (margin 597), and all constants decrease with v . All numbers are ball-arithmetic enclosures. $\square$

6. THE CENTRE REGIME $\mu \geq 0.098$

Here $c_n(m) = \frac{1}{2\pi i} \oint P_n(q)^3 q^{-m-1} dq$ is evaluated directly. Put $L(u) = 3 \sum_{k \leq 3n, 3 \nmid k} \log(1 - e^{ku})$, $u = \log q$. The dominant contributions come from complex saddles $u^* = \frac{2\pi i}{3} + x^*$ and its conjugate, with $L'(u^*) = m$. Write $\xi = -N \operatorname{Re} x^*$ and $\eta = N \operatorname{Im} x^*$.

6.1. Main term and error budget. The contour is $|q| = e^{\operatorname{Re} u^*}$. With $A = |L''(u^*)|$, $\sigma = A^{-1/2}$, $b = -\frac{1}{2} \arg L''(u^*)$, the main term is $2 \operatorname{Re}[e^{L(u^*)-mu^*} \omega i e^{ib} \sqrt{2\pi/A}/(2\pi i)]$, whose sign is that of $\cos(\text{phase})$ with

$$\text{phase} = \alpha(r) - \frac{2\pi}{3} + b + E_1, \quad \alpha(r) = \arg P_n(r\omega)^3 = -\frac{\pi}{6} + \arctan \frac{\sqrt{3}r^{3n}}{2 + r^{3n}},$$

where $|E_1| \leq \sup |L''| \delta^2/2$ and $\delta = \eta/N$ is bounded by Abel summation from $\operatorname{Im} L' = 0$. The error relative to the main term is $(2(\text{inner} + \text{middle}) + \text{far})/(2|\cos \text{phase}|)$, where: *inner* is the Taylor window $|y| \leq c\sigma$ (third-derivative constant $G_3 = H_3\sigma^3$, $H_3 = \sup_{\text{window}} 3 \sum k^3 |z|(1 +$

$|z|)/|1-z|^3$ with $|z|$ and $|1-z|$ enclosed as real balls); *middle* is $c\sigma \leq |\theta - \theta^*| \leq 0.05$, bounded in three zones in $t = N(\theta - \frac{2\pi}{3})$: $|t| \leq 0.8$ by the curvature $\operatorname{Re} L'' \geq N^3 \rho_{\text{lo}}(t) - 3N^2 V$, $0.8 \leq t \leq 10$ by the linearised continuum profile $-N \int_0^1 g(e^{-\xi\tau})(1 - \cos \tau t) d\tau$ plus a Koksma remainder, and $t \geq 10$ by the Dirichlet kernel; *far* is $|\theta - \frac{2\pi}{3}| \geq 0.05$, bounded by $G \leq -\frac{3}{2} \sum g_k(2 \cos k\theta + 1)$ with a Koksma profile for $N\theta \leq 40$ and the Dirichlet kernel beyond.

6.2. Uniformity in N . In scaled variables every piece is an explicit function of (ξ, t, ϵ) whose N -dependence is either through Koksma/Cauchy remainders of order $1/N$ with the favourable sign, or through factors $N^p e^{-cN}$ with an asserted slope. The tables `midu_table`, `g3u_table`, `clowu_table` record upper bounds valid for all $N \geq 733$ on 110 balls covering $\xi \in [-0.3, 3.4]$; the shift margin for η is 0.01, 0.015, 0.02 on $[-0.3, 1.75]$, $[1.75, 3.05]$, $[3.05, 3.4]$ and exceeds the certified $|\eta| \leq 0.0155$.

Certificate 6.1 (`coverage.py`, check E). On every one of the 110 ξ -balls the error ratio computed from the tables alone is below 0.4905, and $\cos(\text{phase}) < 0$.

Remark 6.2 (existence of the saddle). For each $N \geq 733$ and $\mu \in [0.098, \frac{1}{2}]$ the saddle u^* exists and is unique in the region $|N \operatorname{Im}(u - \frac{2\pi i}{3})| \leq 0.02$: the real equation is solved by monotonicity of $\mu_N(\xi)$, and the imaginary offset by a Newton–Kantorovich step using the tabulated lower bound for $\operatorname{Re} L''$ and the window bound for $|L'''|$ (the same constants that bound η).

Lemma 6.3 (`coverage.py`, check F). For all $N \geq 733$: $\mu \geq 0.098 \Rightarrow \xi \leq 3.4$ and $\mu \leq \frac{1}{2} \Rightarrow \xi \geq -0.3$.

Proof. $\mu_N(\xi) = N^{-2} \operatorname{Re} L'(u^*)$ equals $\int_0^1 \tau \sum_j \operatorname{Re} \frac{-z_j}{1-z_j} d\tau$ up to a Koksma error $3 \sum_j (\operatorname{TV}_j + S_j)/N$; it is decreasing in ξ since $\operatorname{Re} L'' > 0$. Enclosures give $\mu_N(3.4) \leq 0.0886$ and $\mu_N(-0.3) \geq 0.531$. $\square$

7. PROOF OF THEOREM 1.2

For $n \leq 243$ use Certificate 3.1. Let $n \geq 244$ and $m \equiv 2$, $m \leq \frac{9}{2}n^2$. If $m < 3n$ then $c_n(m) = 0$ (all exponents of P_n^3 below $3n + 1$ except $0, 1, 2, \dots$ come from P_∞^3 , whose class-2 coefficients vanish). If $3n \leq m < 6n + 2$ use Proposition 4.1. If $m \geq 6n + 2 = 2N$ and $\tilde{\mu} < 0.0985$, the saddle v satisfies $v > 2.95$ (Certificate 5.1); use Certificate 5.1 for $v \leq 3000$ and Proposition 5.2 for $v \geq 3000$. Otherwise $\tilde{\mu} \geq 0.0985$, so $\mu = \tilde{\mu} + (N + \frac{1}{4})/N^2 \geq 0.098$, and $\mu \leq \frac{1}{2}$; by Lemma 6.3 the saddle lies in the tabulated range and Certificate 6.1 applies. $\square$

8. COMPUTATIONS AND REPRODUCIBILITY

All scripts and logs are in the accompanying repository; `run_all.sh` regenerates everything from scratch (about two hours on one core). Software: Python 3.12, python-flint 0.9.0 (Arb), mpmath 1.3.0, numpy 2.4, GMP 6.3.

script	certifies	result
<code>exact/cubic_check.c</code>	Cert. 3.1	all $n \leq 243$
<code>band_certificate.py</code>	Cert. 4.2	exact $i \leq 2000$; tail ratio $< 10^{-33}$
<code>centre*_uniform*.py</code>	centre tables	110 ξ -balls
<code>run_uniform.py</code> , <code>B_uniform.py</code>	Cert. 5.1	36 chunks, 0 unresolved, worst 0.117
<code>ray_asymptotic2.py</code>	Prop. 5.2	ratio ≤ 0.0082
<code>coverage.py</code>	Cert. 6.1, Lemma 6.3, tiling	COVERAGE COMPLETE
<code>poisson_check.py</code>	Lemma 2.4	$ \eta \leq 1.42$

9. INDEPENDENT CHECKS (NOT USED IN THE PROOF)

- **identity_check.py**: $c_n(m) = -2(\text{Im } J_0 + \text{Im } J_\eta) + R(m)$ with every term computed independently; relative error $\leq 7 \cdot 10^{-9}$ ($n = 12$) and $\leq 6 \cdot 10^{-10}$ ($n = 20$).
- **expansion_check.py**: $|\rho(i)|/(\text{bound}) \leq \sqrt{3}/2$ for all $i \leq 3000$ and 3388 values $i \leq 108000$, exactly.
- **exact_accept.py**, **centre_accept.py**: exact $c_{244}(m)$ divided by the Laplace, resp. centre, main term is 0.9934–0.9995, resp. 0.9995.
- **B_exact.py**: an independent certificate at $n = 244$ for $\mu \leq 0.2$ using the unscaled closed forms (no failures).
- **hand_checks.py**, **koksma_cauchy_check.py**, **sz_expansion_audit.py**: symbolic check of the midpoint Peano kernel, randomised checks of the Cauchy and Koksma inequalities, numerical audit of every constant in Appendix A.

APPENDIX A. THE EXPLICIT EXPANSION OF p_3

Put $G(\tau) = \prod_{3|m} (1 - e^{2\pi i m \tau})$, so $p_3(i) = [e^{2\pi i i \tau}]G^3$. With $f(x) = \prod_{m \geq 1} (1 - x^m)^{-1}$ and the eta transformation, for $\tau = \frac{h}{k} + \frac{iz}{k^2}$, $\text{Re } z > 0$, $d = \gcd(3, k)$ and $hh' \equiv -1 \pmod{k}$:

$$G(\tau)^3 = \left(\frac{3}{d}\right)^{3/2} e^{i\theta_{h,k}} \exp\left(\frac{3\pi(d^2 - 3)}{36z} - \frac{\pi z}{2k^2}\right) \widehat{G}^3, \quad \widehat{G} = \frac{f(e^{2\pi i d h'_d/k - 2\pi d^2/(3z)})}{f(e^{2\pi i h'/k - 2\pi/z})},$$

with a real $\theta_{h,k}$.

- (1) If $3 \mid k$ ($d = 3$) the exponential is $e^{\pi/(2z)}$ (so $B = \frac{1}{2}$) and $\widehat{G}^3 = 1 + \sum_{j \geq 1} \varphi_j e^{-2\pi j w}$ -type with majorant $f(t)^3 f(t^3)^3$, $t = e^{-2\pi \text{Re}(1/z)}$. The correction term would grow only if $3 > 24/(3 - 1) = 12$; it does not.
- (2) If $3 \nmid k$ the exponential decays and $|G^3| \leq K_3^3$ on $\text{Re}(1/z) \geq 1$, $K_3 = \sqrt{3}e^{-\pi/18} f(e^{-2\pi/3}) f(e^{-2\pi}) = 1.691256$ (the maximum is at $\text{Re}(1/z) = 1$).

Dissect $[0, 1)$ by Ford circles of order $N \geq \sqrt{4\pi i} + 1$ and map each to the circle $K : |z - \frac{1}{2}| = \frac{1}{2}$. On the chords and the connecting arcs $|e^{\pi(2i - \frac{1}{2})z/k^2}| \leq e$. For $3 \mid k$, complete the main integral to all of K : $\frac{i}{k^2} \int_K e^{\pi B/z + \pi(2i - B)z/k^2} dz = \mathcal{P}_k(i)$, at the cost of the two arcs, each of length $\leq \frac{\pi}{2} \sqrt{2k}/(N+1)$; move the remainder to the chord, where the majorant is non-increasing in $\text{Re}(1/z)$ because $B \leq 2$. For $3 \nmid k$ move to the chord and use K_3^3 . Counting $\sum_{3|k \leq N} \varphi(k)/k \leq \frac{2}{9}(N+1)$ (as $\varphi(3j)/3j \leq \frac{2}{3}$) and $\sum_{k \leq N} \varphi(k)/k \leq N$ gives $|\rho(i)| - \sum_{3|k \geq 6} \varphi(k) \mathcal{P}_k \leq E_{\text{arc}} + E_{\text{rem}} + E_{\text{ng}}$ with

$$E_{\text{arc}} = \sqrt{2}\pi \frac{2}{9} e^{1+\pi/2} = 12.9103, \quad E_{\text{rem}} = 2\sqrt{2} \frac{2}{9} e^{1+\pi/2} (f(t)^3 f(t^3)^3 - 1) = 0.0463, \quad E_{\text{ng}} = 2\sqrt{2} e K_3^3 = 37.1935,$$

$t = e^{-2\pi}$, total 50.1501. The $k = 3$ amplitude is $\sqrt{3}\chi(i)$, obtained from the Dedekind-sum phases and confirmed exactly for $i \leq 108000$ (§9).

APPENDIX B. AUXILIARY INEQUALITIES

(i) *Cauchy*: if f is holomorphic on $|\zeta - c| \leq \rho$ with $|f - f(c)| \leq M$ on the boundary, then $|f^{(k)}(p)| \leq k! M \rho / (\rho - d)^{k+1}$ for $|p - c| \leq d < \rho$. (ii) *Midpoint*: $|g(0) - \int_{-1/2}^{1/2} g| \leq \frac{1}{8} \int |g''|$ (Peano kernel $(\frac{1}{2} - |t|)^2/2$). (iii) *Koksma*: one node per cell of width h gives $|h \sum f(x_i) - \int f| \leq h \text{TV}(f) + (\text{uncovered length}) \sup |f|$. (iv) *Envelopes*: $\sqrt{2\pi x} e^{-x} I_\nu(x)$ is increasing to 1 for $\nu \geq \frac{1}{2}$.

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